A Tool-Using AI Agent Forecast of the S&P 500 Close

Factor search on the group dataset, a pre-registered selection rule, chronological validation and the agent workflow behind one number

Group	9
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Build	FINAL run, generated 2026-11-13 14:47 UTC from the saved snapshot
Main-body words	2,321 of 2,400 allowed (excludes cover, references, declaration, contribution statement, appendix)

BINDING FINAL POINT FORECAST

7,637.84

S&P 500 price index, index points

Market / unit	S&P 500 price index (SPX), index points
Target	Official closing price-index level on Fri 20 Nov 2026, 16:00 America/New_York (21:00 UTC)
Information cutoff	Fri 13 Nov 2026 12:00 UTC. Last usable close: Thu 12 Nov 2026 = 7,592.40
Reference source	S&P Dow Jones Indices official closing price-index level (not futures, ETF or total-return series)
80% prediction interval	7,405.94 to 7,816.57 (not scored)
Discretionary adjustment	0.00% - none applied

1. Executive summary and final forecast

Forecast: 7,637.84 index points for the S&P 500 official close on Fri 20 Nov 2026: 7,592.40 (last close, Thu 12 Nov 2026) x exp(+0.40% expected 6-session return and +0.20% overnight futures nowcast), with no discretionary adjustment. The 80% interval is 7,405.94 to 7,816.57 (+/-2.7%, VIX 18.7); only the point forecast is scored.

- **Factor search, then selection by rule.** Of 16 pre-registered candidates, we screened the 12 computable from public data, 6 macro and cross-asset (yields, USD, gold, semiconductors), each against median drift on a validation block; the 4 that need the workbook were tested separately and failed. **None passed.** The best, 2Y yield 20-day change (validation t 1.53), misses the bar. The model is frozen: no factor can enter on submission day.
- **Median drift is no better than unchanged.** Test-block error 1.909% against 1.923% (243 origins; Diebold-Mariano t = 0.53). The best-ranked candidate alone scored 1.925%.
- **News enters through prices, not opinion.** The 07:00 New York futures price reflects the latest data, Fed and earnings news. In the group's pre-registered test on Bloomberg prices it cut six-session error from 1.389% to 1.307% (133 origins, t 2.50); a second test over 930 origins finds the move carries through about one for one (slope 0.92).
- **Accuracy is mostly luck.** Typical error is 1.4% (score about 40.5/50); we claim a well-centred, reproducible number, not foresight.

2. Market, data and information boundary

Market. The S&P 500 *price* index, scored on its official close on Fri 20 Nov 2026. The cutoff, Fri 13 Nov 2026 12:00 UTC (07:00 New York), precedes the US open, so the last official close in hand is Thu 12 Nov 2026; the horizon is h = 6 NYSE sessions. A value enters only if its official close time is at or before the cutoff; intraday bars only once finished. A leak test corrupted 29 post-cutoff observations across all input files; model inputs did not change.

Table 1. Observed data and timing discipline

Source (raw files saved, SHA-256 in Appendix A)	Publication time	Retrieved	Last usable observation
S&P 500 and VIX daily, Yahoo (history from 1990; the model input)	official close 16:00 New York (VIX 16:15)	2026-11-13 14:29 UTC	12 Nov 2026 close
E-mini S&P 500 5-minute bars, Yahoo (nowcast)	continuous; a bar is used only once it has ended	2026-11-13 14:29 UTC	bar ending 13 Nov 12:00 UTC
US Treasury par yields (official), gap fill and check	after the close; used from 17:00 New York	2026-11-13 14:30 UTC	latest published row
Cboe SPX and VIX history (independent check)	after the close	2026-11-13 14:29 UTC	reconciliation only

Reconciliation. Yahoo and Cboe SPX differ by over 0.05% on 16 of 9,284 days (largest: a stale Cboe value). 3 Yahoo values confirmed wrong by three sources are replaced with Cboe's (price_corrections.json); model inputs are public data only. Human sign-off checks the last close against S&P DJI.

3. Latest news and market regime

Researched on 2026-11-10 from dated news pages and checked against saved prices ([data]). October ended with an AI-spending scare and a Federal Reserve leaning toward tightening; on the latest information the S&P 500 sits 2.9% below its one-year high, 21-day realised volatility is 22% and VIX 18.7. Risk is two-sided: hawkish Fed and AI-capex doubts against sharp relief rallies.

Table 2. Dated events (2026)

Date	Event	Market read	Source
11/04	US midterms settle; markets cite removal of election uncertainty (sources differ on Senate control).	S&P 500 +1.6% to 7,688.62 [data].	Russell Investments
11/05	AI and chip rout: Nvidia about -8%, AMD about -16%, Nasdaq about -5.5% (correction).	S&P 500 about -4.4% to 7,353.73 [data], worst day since April 2025; VIX 28.18 [data].	CNBC
11/06	October payrolls +165k against 65k expected; December hike odds rise to about 78%.	S&P 500 +0.31% to 7,376.86 [data]; week about -2.7%.	BBN Times
11/09	Nvidia board opens an independent probe linked to its CEO; shares about -4%.	S&P 500 +2.35% to 7,549.94 [data], strongest session since April.	CNBC
11/10	October CPI +0.4% (0.3% expected); core +0.3% (0.2% expected), third month at that pace. Waller floats a 50bp December hike.	S&P futures about -0.5% pre-market; 10Y Treasury 4.94% (Yahoo ^TNX).	CNBC

How news enters the forecast. Known news is already in the 07:00 New York futures price, which the nowcast uses; events after the cutoff (Nvidia results 17 Nov, FOMC minutes 18 Nov, options expiry 20 Nov) widen the interval instead. No hand adjustment: 0.00%.

4. Factor research and economic mechanism

Protocol (fixed before results). Each candidate has an economic sign chosen in advance. For each, a walk-forward median (LAD) regression forecasts the deviation of the 6-session return from median drift; weekly origins from 2018-01-01 (204 validation origins to 2021-12-31, then 243 test origins). It is eligible only if its fitted sign matches, the Diebold-Mariano t of its error against drift is at least 1.65 and a 2000-2012 long-history proxy does not contradict it (without a proxy: 2.33). With 16 tests about one false positive is expected at 1.65, hence the stricter bar.

Table 3. Factor screen (t > 0 means better than median drift; test t is reported, not used)

Candidate factor	Mechanism	Sign	t valid	t long	t test	Result
2Y yield 20-day change	policy tightening	-	1.53	n/a	-0.62	fails
VIX level (fear premium)	fear premium	+	0.96	0.10	0.15	fails
USD 20-day return	foreign earnings hit	-	0.41	-0.53	0.97	fails
Drawdown from 1-year high	rebound after falls	-	-0.04	-2.93	0.57	fails
5-day reversal	short-term overreaction	-	-0.62	-0.49	-0.48	fails
5-day VIX change	fear spikes fade	-	-0.76	0.41	-1.17	fails
252-day trend	trends persist	+	-0.80	-1.13	-1.99	fails
10Y yield 20-day change	discount rate rises	-	-0.96	0.86	-0.27	fails
Variance risk premium	insurance premium earned	+	-1.07	0.28	-1.27	fails
Semiconductors vs S&P 500	AI leadership	+	-1.26	-1.14	-2.10	fails
Gold 20-day return	risk-off flight	-	-1.52	0.69	-0.87	fails
Curve slope 10Y-2Y	growth outlook	+	-1.62	-1.49	-2.07	fails

Result: no candidate beyond the two model factors survives. Over 6 sessions none reliably beat drift out of sample, in line with Welch and Goyal and with Campbell and Thompson. The group agrees: 30 pre-registered tests (hurdle t 3.14) passed none, and the 1.72 million-combination search's best model lost to plain drift on unseen data. 4 further candidates need licensed Bloomberg data (economic surprise level, economic surprise 20-day change, cpi inflation change (3m), ppi inflation change (3m)); tested separately on the group's Bloomberg workbook, none passed (validation t -1.51 to -0.45).

Table 4. The two model factors as computable variables

Specification	D: equity risk premium (median drift)	N: overnight futures move
Formula	median of ln(S[t+6]/S[t]) over all realised windows	ln(ES at cutoff / ES at 16:00 New York close) minus its average
Source	S&P 500 official close (Yahoo, Cboe-checked)	CME E-mini S&P 500 front month, 5-minute bars (Yahoo)
Publication time	16:00 New York, each session	continuous; read at the cutoff (07:00 New York)
Frequency / lookback	daily; expanding window since 1990	intraday; since the last cash close
Expected sign	+ (paid for bearing equity risk)	+, one for one (group test slope 0.92)
Mechanism	risk premium; negative skew lifts the median above zero	futures lead index price discovery [Hasbrouck] and price overnight news; arbitrage ties the cash open to them
Fails when	prolonged bear markets (2000-02, 2008)	contract roll, thin overnight trading, reversal at the open

Why these two. The last close prices news up to 16:00 New York; E-mini futures trade overnight, so their move to 07:00 prices the news since. Beyond that, six-session returns have no reliably forecastable part, leaving the equity premium (drift). It fails on a reversal at the open, a contract roll or thin overnight trading.

5. Agent architecture and modelling

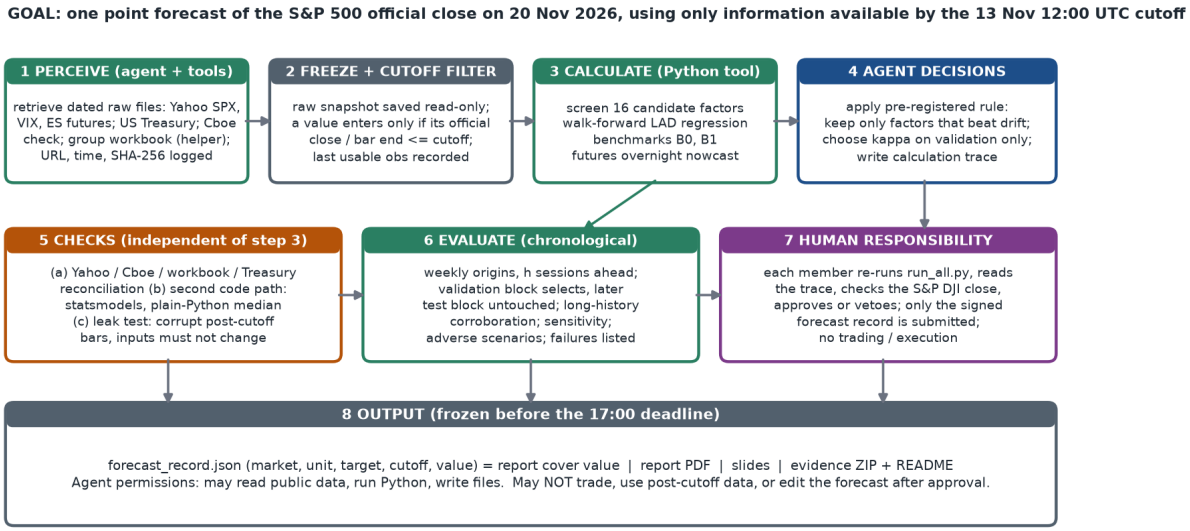


Figure 1. Labelled agent workflow: goal, data, tools, agent decisions, checks and human responsibility.

Agent. Claude Code (Claude Sonnet 5.5) may read public data, run Python and write files; it cannot trade or submit. It retrieved the raw files, called Python to screen factors and fit models, applied the pre-registered rule and logged each decision in agent_log.md. Members re-run the code, read the trace and approve the record. The agent's own check of the record (plan, tool calls, verdict) is saved in agent/demo_runs/.

- *Decision:* fit by least absolute deviations, because score is linear in absolute percentage error and its optimal point forecast is the conditional median [Gneiting].
- *Decision (group review, 10 Nov):* freeze the model. The screen is reported but cannot change the forecast on submission day (kappa = 0), so the submitted method is exactly the reviewed one.
- *Correction:* reconciliation exposed a stale Cboe value and a faulty Yahoo value; a first solver was too slow and was replaced; a second code path (plain-Python median and nowcast, statsmodels QuantReg) reproduced the forecast (PASS_ALL = True).

Table 5. From baseline to forecast level

Step	Value	Type
Last official close S_t	7,592.40	Observed
Median 6-session return m0 (all history)	+0.400%	Calculated
Expected return = m0 + kappa(0.0) x factor fit	+0.400%	Calculated
Futures nowcast: ES 7,616.25 to 7,635.50 (+0.252%) less mean drift +0.056%	+0.196%	Observed, calculated
Discretionary adjustment	0.00%	Judgement (none)
Forecast = 7,592.40 x exp(+0.597%)	**7,637.84**	Result

6. Validation and benchmark

Design. Weekly origins on the final origin's weekday, 2018-01-04 to 2026-10-29 (447), h = 6; parameters use only rows whose target was already realised. Benchmarks: B0 unchanged, B1 median drift. Loss: score = 50 - min(50, 5 x APE). Price factors train on up to 36 years, macro series on the workbook history (from 2015).

Table 6. Chronological out-of-sample errors

Model	Test MAPE % (n=243)	Validation MAPE % (n=204)	Test mean score /50	Worst test APE %	DM t vs B0 (test)
B0 unchanged	1.923	1.907	40.43	12.2	-
B1 median drift	1.909	1.835	40.51	12.7	0.53
Best-ranked candidate alone	1.925	1.822	40.43	12.8	-0.05
Trend + VRP (earlier design)	1.915	1.877	40.48	12.6	0.31
Final: drift (+ nowcast, validated below)	1.909	1.835	40.51	12.7	0.53

Reading the table. Drift is no better than unchanged (test 1.909% vs 1.923%; validation 1.835% vs 1.907%): a small, regime-dependent edge. The best-ranked candidate (1.925%) and the earlier trend-plus-VRP design (1.915%) do not earn a place. The worst test errors were 20 Feb 2020 (14.6%); 27 Mar 2025 (12.7%); 05 Mar 2020 (12.0%), all shocks the unchanged benchmark missed equally.

Futures evidence (group tests). A pre-registered test on Bloomberg 07:00 New York futures prices (133 origins, Apr-Oct 2026) cut six-session error from 1.389% (drift alone) to 1.307% (t 2.50, below its pre-set hurdle of 3.16, adopted on the pre-set rule of a gain in both halves; Thursdays 1.543% to 1.357%, t 3.49). A second test over 930 origins (Jan 2023-Oct 2026) finds a slope of 0.92 (s.e. 0.16), supporting a one-for-one entry. The group's like-for-like review scored this exact rule at 187 Friday 07:00 New York origins (Jan 2023-Oct 2026): 1.525% against 1.556% for drift alone and 1.603% for no change; after 19 Aug 2025 (55 origins) 1.349% against 1.460% and 1.481% (reality-check p 0.074). Our own backtest of the exact model on the same prices agrees: all 135 origins 1.378% to 1.318% (t 1.68), Thursdays 1.528% to 1.382% (t 2.26). Over 30 years Asian markets' overnight move explains 4.3% of the next US day (7,510 days), the spillover documented by [Hamao], but adds nothing once futures are known.

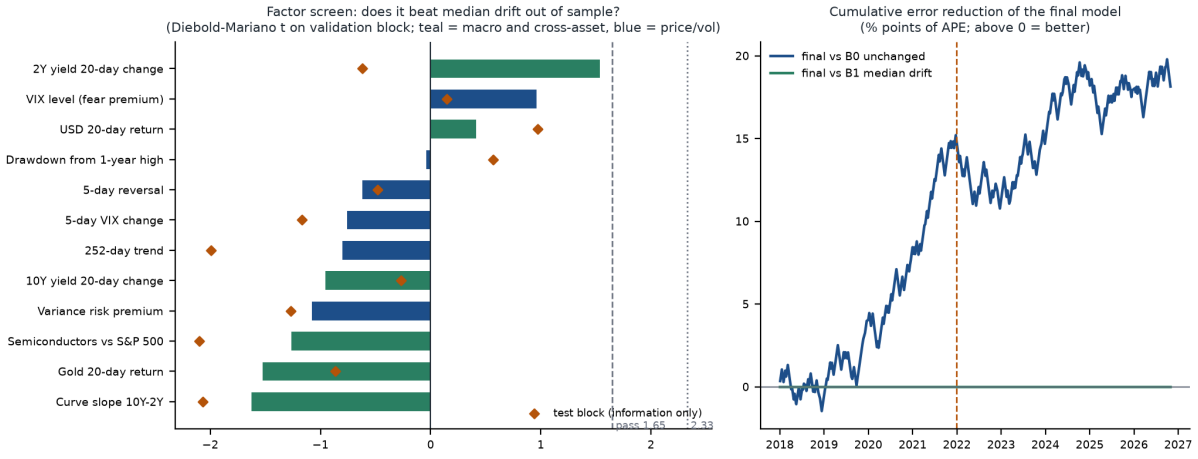


Figure 2. Left: factor screen (validation t; diamonds show the test block). Right: cumulative error reduction of the final model.

7. Sensitivity and adverse scenarios

Table 7. Sensitivity of the final forecast to the drift choice and the futures timing

Assumption	Drift	Nowcast	Forecast	vs final
Final: median since 1990	+0.40%	+0.20%	7,637.84	0.00
Median since 2000	+0.39%	+0.20%	7,636.97	-0.87
Fixed +0.25% per 6 sessions	+0.25%	+0.20%	7,626.36	-11.48
Mean since 1990	+0.20%	+0.20%	7,622.34	-15.50
No drift (unchanged centre)	+0.00%	+0.20%	7,607.31	-30.53
Futures read at 11:00 UTC (06:00 New York)	+0.40%	+0.12%	7,631.72	-6.12
Futures read at 10:00 UTC (05:00 New York)	+0.40%	+0.03%	7,625.41	-12.43
No futures nowcast	+0.40%	+0.00%	7,622.86	-14.98

Across drift choices the forecast ranges 7,607.3 to 7,637.8 (0.40% of the level), well inside the typical error. Reading the futures one or two hours earlier moves it by up to 12.4 points; dropping the nowcast moves it by 15.0. The key assumptions are the drift and the time the futures price is read.

Table 8. Scenario analysis

Scenario at the cutoff (labelled scenarios, not forecasts)	Index move	VIX	Model level	vs forecast
Relief: index +2%, VIX 17, 2Y -10bp, semis +5%	+2.0%	17.0	7,777	1.8%
Adverse A: AI/chip rout continues (index -4%, VIX 30, semis -10%)	-4.0%	30.0	7,324	-4.1%
Adverse B: hawkish shock (index -3%, VIX 27, 2Y +20bp, 10Y +15bp, USD +1.5%)	-3.0%	27.0	7,398	-3.1%

Adverse case. The model has no crash protection: a further chip rout or a hawkish shock moves the forecast one-for-one with the index. The interval's lower end (7,405.94) is 3.0% below the forecast, an error costing 15.2 marks; a 4% move, the 5 Nov size, costs 20.

8. Limitations and counterarguments

- **Counterargument: 'a good analyst would use the news.'** We tested its measurable parts (yields, dollar, semiconductors, surprises, inflation): none beat drift, and the rest is in the futures price. A hand adjustment could not be validated and would add variance.
- **Small samples.** The screen has 204 validation and 243 test weeks, so a true weak factor could be missed (low power); the proxy requirement and the 2.33 bar guard only against accepting a lucky one.
- **Drift is not a proven edge** (1.835% vs 1.907% on validation); in a falling week the +0.40% centre costs about 2.0 marks versus unchanged.
- **Futures evidence is recent and event risk is large.** The Bloomberg test covers Apr-Oct 2026 and the longer one Jan 2023-Oct 2026; Nvidia results, FOMC minutes and options expiry fall after the cutoff.
- **Data risk.** Yahoo is a secondary source, so the last close must be checked against S&P DJI. Nothing here is investment advice.

Contribution statement

Contributions as agreed by the group. The Bloomberg export used for data cross-checks and the exploratory macro screen is not a forecast input and is not in the evidence ZIP.

Member	Responsibilities	Checks performed and approved
Layla Ahmed, 26236281	Submission pipeline: data snapshot and cutoff filter, 16-factor screen, futures nowcast, validation, report, slides, evidence ZIP	Independent second code path and leak tests; forecast reproduced by Antonio and Jayen (7,543.72 for the 13:45 UTC preview of 10 Nov, 7,598.08 for the 12:00 UTC preview)
Antonio Herrera Despradel, 26051351	Pre-registered drift + Asia model (30-test budget), Bloomberg export (UCL terminal), futures test on 07:00 prices, review of all members' work	Placebo and sub-period tests; reproduced Layla's package; compared all models like for like
Jayen Rughani, 26150296	1.72-million-combination factor search and drift experiments; pre-registered semiconductor test and ES-futures 07:00 test on TradingView data (PR #5); review and ranking of all members' models on common origins (PR #8); group repository, Windows data-hash fix and submission check	Tests pre-registered before results; 70/30 unseen-window evaluation; Newey-West t, block bootstrap and reality check; 17 data checks; reproduced Layla's forecast from her rule (7,543.69 vs 7,543.72) and from the evidence ZIP on a second machine (7,598.08)
Mohamed Derbala, 26077489	Momentum + RSI machine-learning model (80/20 split), five-session forecasts	Reproducible scripts; robustness note on technical indicators

AI-use declaration

The group used Claude Code (Sonnet 5.5, Opus 5.5 and Haiku 5.5; Claude Fable 5.2 for one repository review) and the Codex desktop assistant (model/version not shown in the chat record). These tools assisted with data preparation, coding, model and factor analysis, checks, and preparation of the report and slides. Antonio also asked ChatGPT, Gemini and Meta AI the same question once; their answers were treated as ideas to test and were not cited.

Layla: Claude Code assisted with the data pipeline, factor screen, futures adjustment, validation, report and slides. The work was checked with an independent code path, leakage tests and forecast reproduction.

Antonio: Claude Code assisted with the drift and Asia model, data and futures tests, and review of the group's work. Checks included pre-registered tests, data comparisons and reproducing the package.

Jayen: Claude Code assisted with factor searches, model experiments, futures tests and comparisons of the group's models. Evaluation used a held-out time period and statistical checks.

Mohamed: Codex assisted with analysis of the S&P 500 data and factors, the momentum-plus-RSI model, backtesting, charts and presentation material. Forecasts were compared with actual values, using a chronological 80/20 split for the machine-learning evaluation.

AI-assisted work was checked against saved outputs and actual data. Timing, data-alignment and futures-roll issues found during testing were corrected; weaker model results were reported as such.

Each member's detailed entry (tools and models, significant instructions, checks, errors corrected) is in group_info.json and the agent's instructions and log are in agent/AGENT_INSTRUCTIONS.md and agent/agent_log.md, all in the evidence ZIP.

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Appendix A. Evidence record (evidence only; no new analysis)

A1. Raw snapshot manifest

File	Role	URL (query omitted)	Retrieved	SHA-256	Bytes
gspc_daily.json	S&P 500 price index daily close (Yahoo Finance chart API, ^GSPC)	https://query1.finance.yahoo.com/v8/finance/chart/%5EGSPC	2026-11-13 14:29:56 UTC	89fb45c3fe0ca8...	1,000,132
vix_daily.json	Cboe VIX daily close (Yahoo, ^VIX)	https://query1.finance.yahoo.com/v8/finance/chart/%5EVIX	2026-11-13 14:29:56 UTC	463463d12f73ac...	974,175
es_5m.json	E-mini S&P 500 front-month future, 5-minute bars (Yahoo, ES=F)	https://query1.finance.yahoo.com/v8/finance/chart/ES%3DF	2026-11-13 14:29:56 UTC	58f59fa003a938...	755,412
tnx_daily.json	10-year Treasury yield, % (Yahoo, ^TNX) - long-history proxy for robustness	https://query1.finance.yahoo.com/v8/finance/chart/%5ETNX	2026-11-13 14:29:56 UTC	b360808e9e0524...	959,432
irx_daily.json	13-week T-bill yield, % (Yahoo, ^IRX) - long-history proxy for robustness	https://query1.finance.yahoo.com/v8/finance/chart/%5EIRX	2026-11-13 14:29:57 UTC	0116d64a2d4dc7...	972,989
dxy_daily.json	US Dollar Index (Yahoo, DX-Y.NYB) - proxy / gap fill for the supporting USD seri	https://query1.finance.yahoo.com/v8/finance/chart/DX-Y.NYB	2026-11-13 14:29:57 UTC	994f5605a91c4f...	1,024,838
gold_daily.json	Gold front-month future (Yahoo, GC=F) - proxy / gap fill for the supporting gold	https://query1.finance.yahoo.com/v8/finance/chart/GC%3DF	2026-11-13 14:29:57 UTC	d2730eae2d49b9...	606,765
sox_daily.json	PHLX Semiconductor index (Yahoo, ^SOX) - proxy / gap fill for the supporting sem	https://query1.finance.yahoo.com/v8/finance/chart/%5ESOX	2026-11-13 14:29:57 UTC	b990992c3d0587...	820,107
n225_daily.json	Nikkei 225 daily close (Yahoo, ^N225) - Asian overnight information	https://query1.finance.yahoo.com/v8/finance/chart/%5EN225	2026-11-13 14:29:57 UTC	f0efdb3f170079...	854,287
hsi_daily.json	Hang Seng daily close (Yahoo, ^HSI) - Asian overnight information	https://query1.finance.yahoo.com/v8/finance/chart/%5EHSI	2026-11-13 14:29:58 UTC	2975ad0db650cb...	846,446
kospi_daily.json	KOSPI daily close (Yahoo, ^KS11) - Asian overnight information	https://query1.finance.yahoo.com/v8/finance/chart/%5EKS11	2026-11-13 14:29:58 UTC	600301b05f1af9...	772,917
asx_daily.json	ASX 200 daily close (Yahoo, ^AXJO) - Asian overnight information	https://query1.finance.yahoo.com/v8/finance/chart/%5EAXJO	2026-11-13 14:29:58 UTC	6f7113cd00604c...	770,784
check_cboe_spx.csv	INDEPENDENT CHECK: Cboe published SPX daily history	https://cdn.cboe.com/api/global/us_indices/daily_prices/SPX_Hi	2026-11-13 14:29:58 UTC	69cf58ab945740...	293,677
check_cboe_vix.csv	INDEPENDENT CHECK: Cboe published VIX daily history	https://cdn.cboe.com/api/global/us_indices/daily_prices/VIX_Hi	2026-11-13 14:29:58 UTC	2a21208b9e9f61...	474,808
treasury_yields.csv	US Treasury daily par yield curve (official); 2 Yr and 10 Yr columns	https://home.treasury.gov/resource-center/data-chart-center/in	2026-11-13 14:30:16 UTC	aa5f6c3fde4ef7...	35,768

A2. Reproduction and versions

Item	Value
Forecast record	market S&P 500 price index; unit index points; target Fri 20 Nov 2026 official close; cutoff Fri 13 Nov 2026 12:00 UTC; value 7,637.84
Software	Python 3.13.2, numpy 2.5.4, pandas 3.1.1, openpyxl, statsmodels (check only), matplotlib, reportlab, python-docx, python-pptx
Independent check	PASS_ALL = True; m0 and nowcast recomputed by plain Python; leak test pass = True; recomputed forecast 7,637.84
Reproduce (no internet or AI account)	python code/run_all.py --final reads the saved files in data/raw_final/ and rewrites outputs, figures, report, slides and forecast_record.json
Supporting workbook	Optional cross-check and macro-screen helper; not a forecast input and not in the submission ZIP (its hash is in the local manifest)
Credentials	None used or stored.